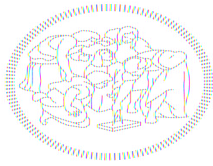




POLITECNICO
MILANO 1863

Bridging Realities: Advancing Human-Avatar Interaction through Sensory Translation Systems in the Physical Metaverse

Tesi di Laurea Magistrale in Computer Science and Engineering - Ingegneria Informatica

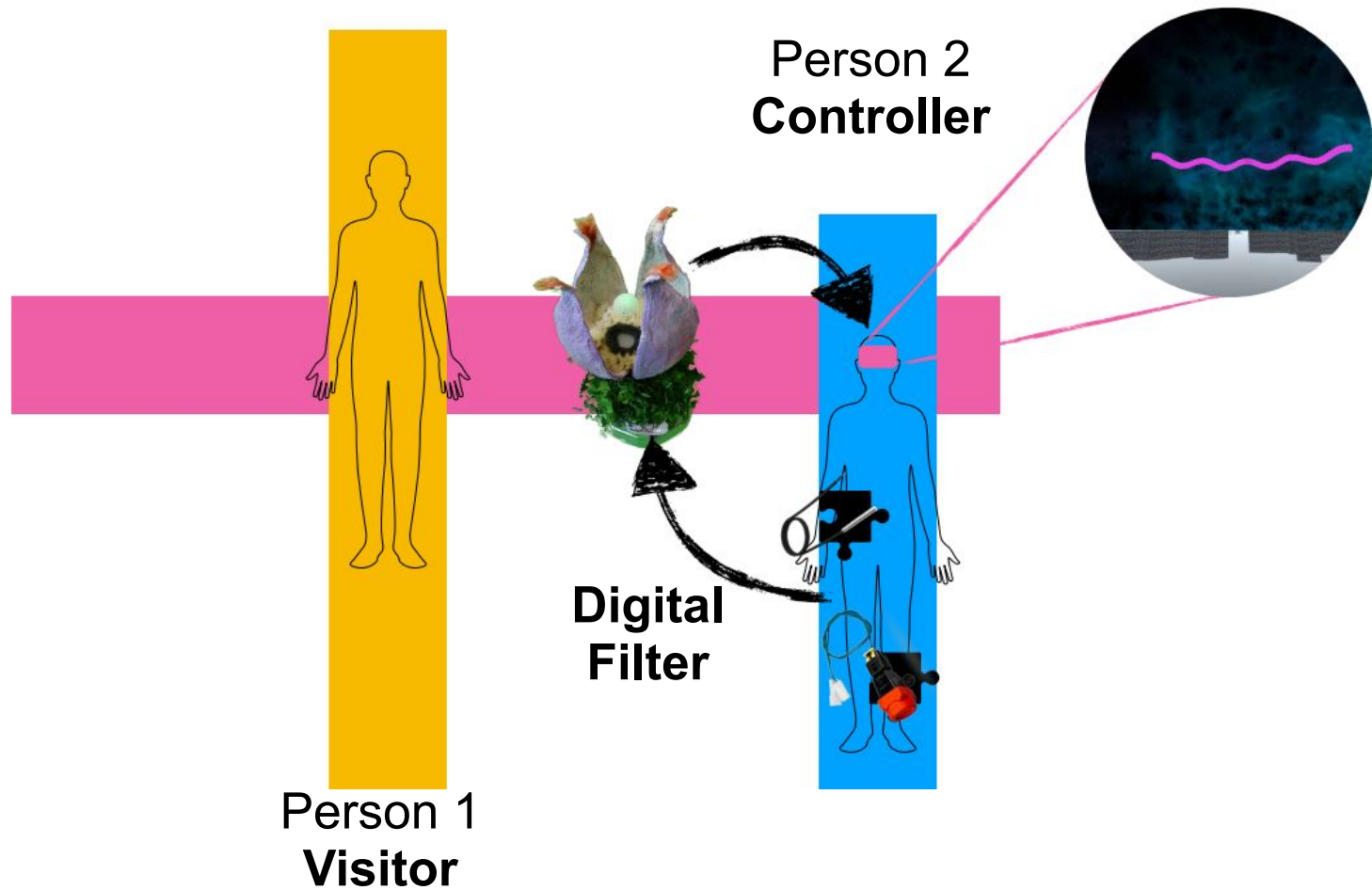


POLITECN
MILANO 1863

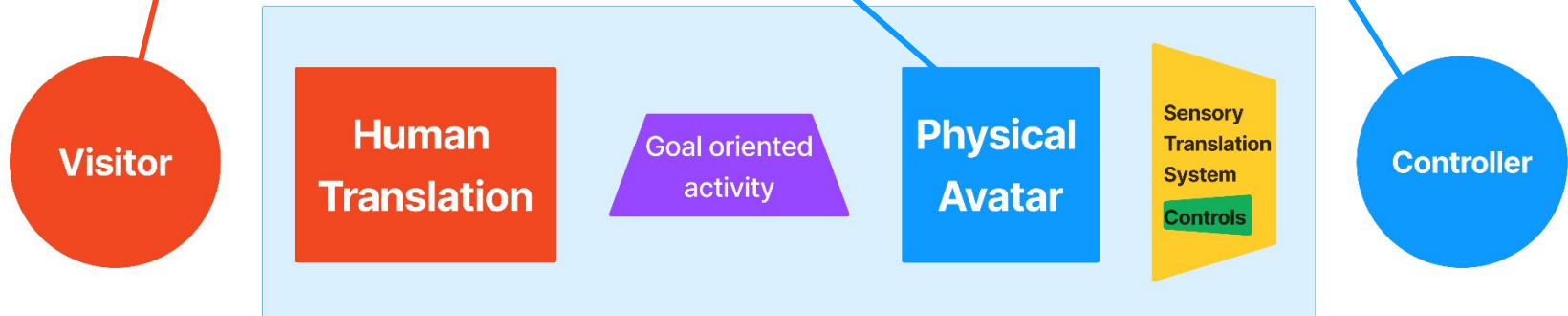
AIRLAB
ARTIFICIAL INTELLIGENCE AND ROBOTICS LAB

Author: Maurizio Vetere
Advisor: Prof. Andrea Bonarini
Co-advisor: Federico Espositi
Academic Year: 2022-23

The Physical Metaverse

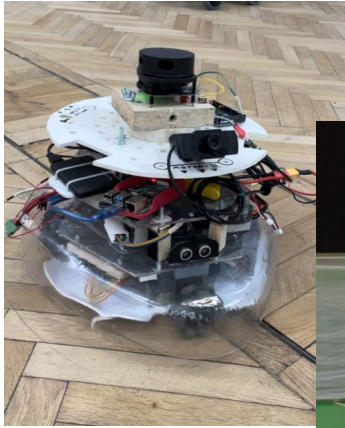


Human-Avatar interaction

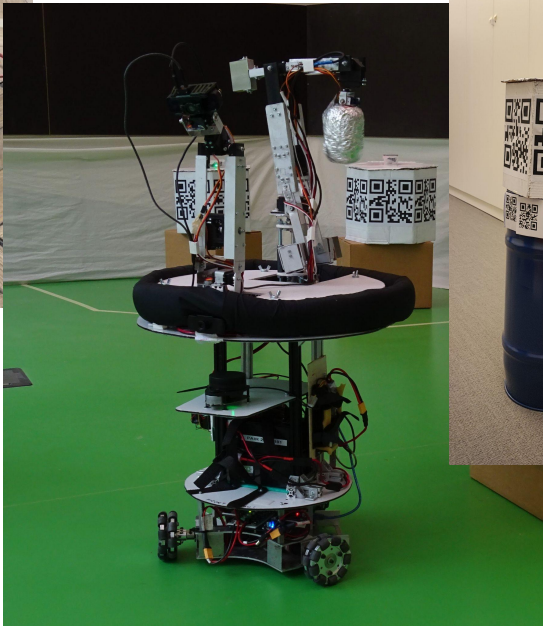


Three different physical avatars

Base



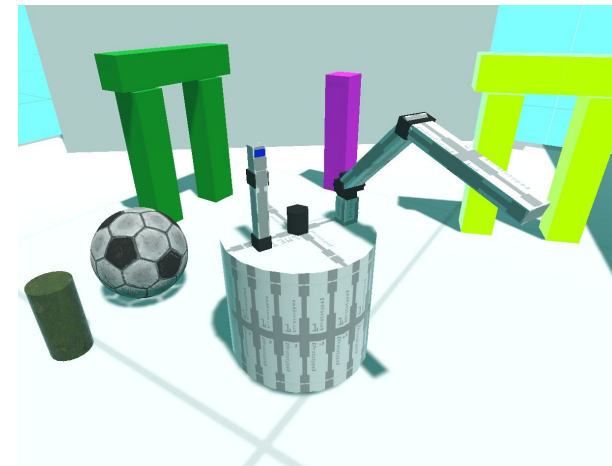
Odile



Blackwings



plus a **virtual** one



Sensory Translation System - Avatar interface

Sensory
Translation
System
Controls

Visitor

Human
Translation

Goal oriented
activity

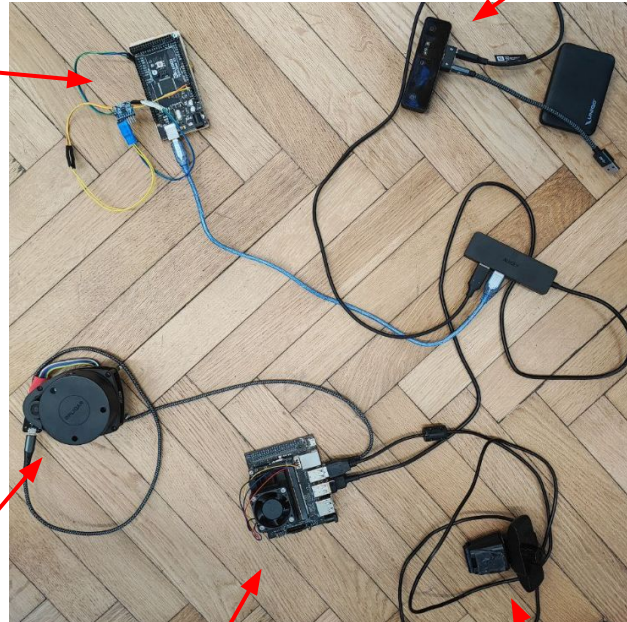
Physical
Avatar

Sensory
Translation
System
Controls

Controller

DepthAI Oak D pro

Arduino



LIDAR

Jetson NANO

Full HD webcam



Sensory Translation System - Controller interface

Sensory
Translation
System
Controls

Visitor

Human
Translation

Goal oriented
activity

Physical
Avatar

Sensory
Translation
System
Controls

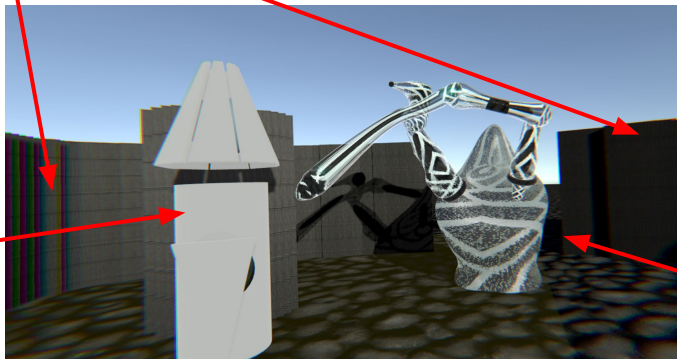
Controller



Oculus Quest 2

Obstacles

Station



Human
Translation

Proprioception

Sensory
Translation
System
Controls

Visitor

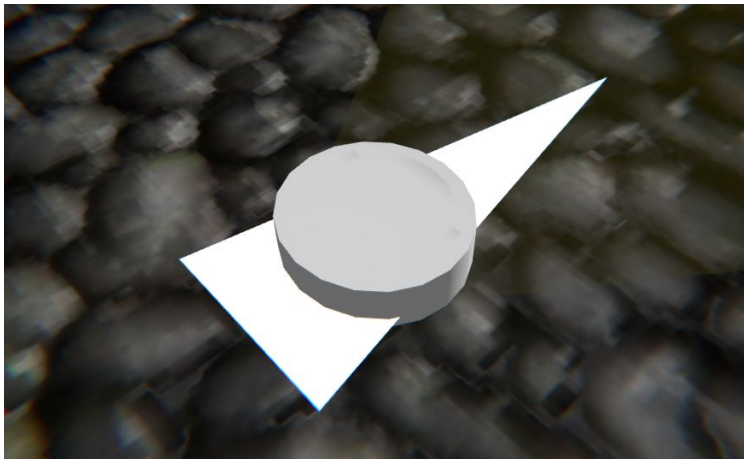
Human
Translation

Goal oriented
activity

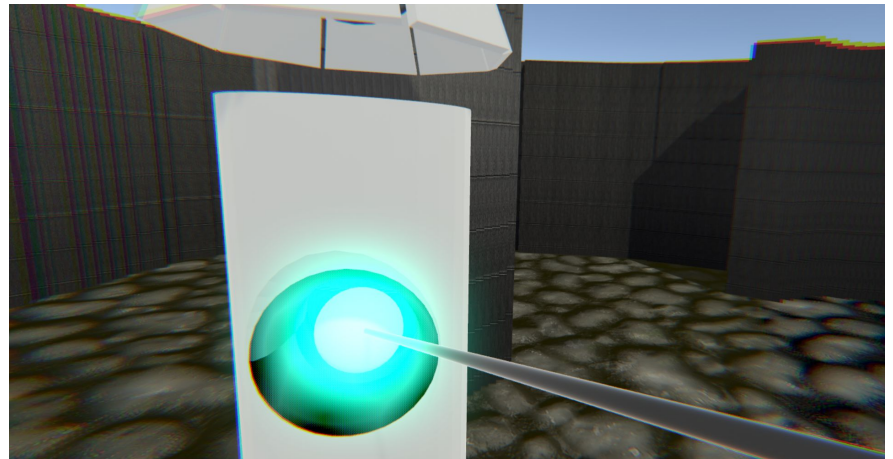
Physical
Avatar

Sensory
Translation
System
Controls

Controller



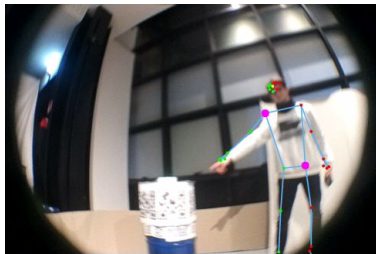
The Controller's **virtual body**



Staff in the user's hand



Qr recognition

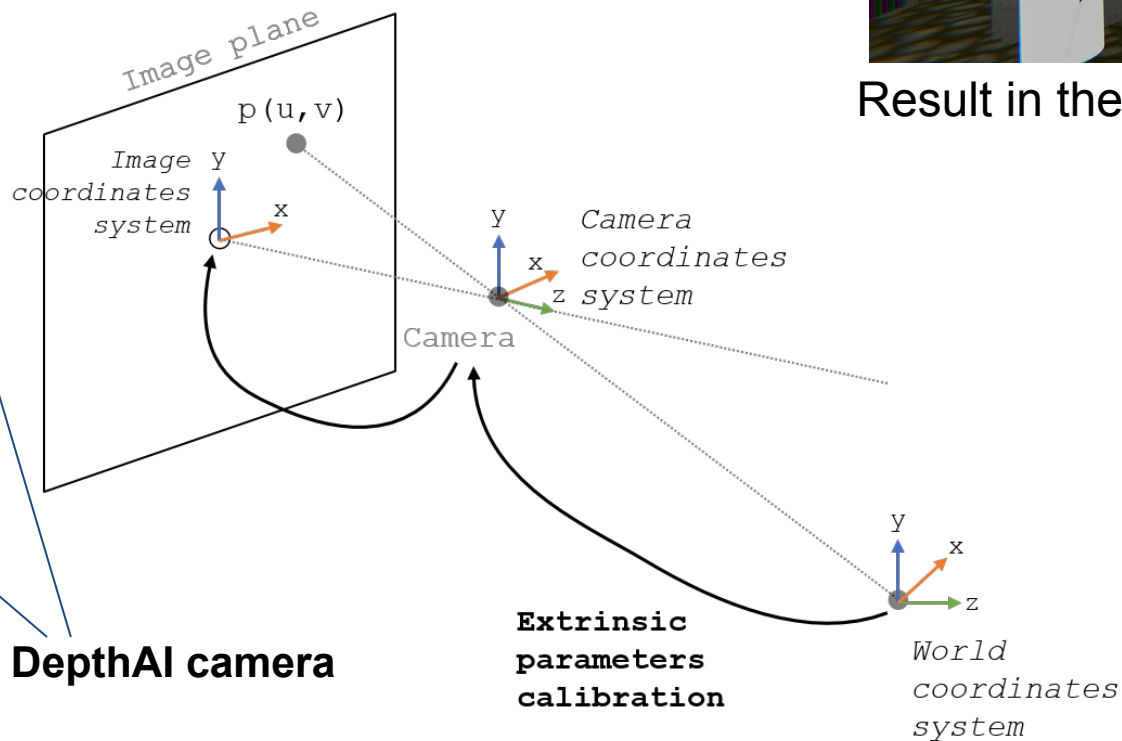


Pose detection

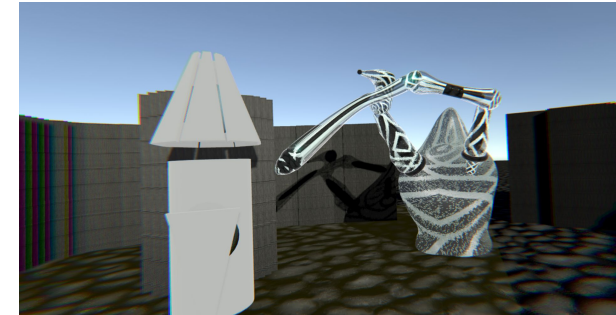


IR depthmap

Webcam



DepthAI camera



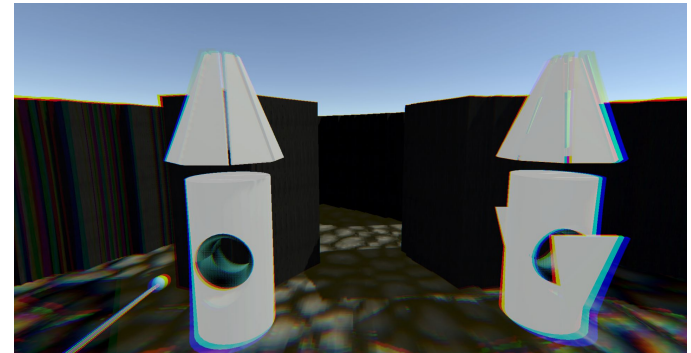
Result in the virtual 3D world

Escape Room-like activity

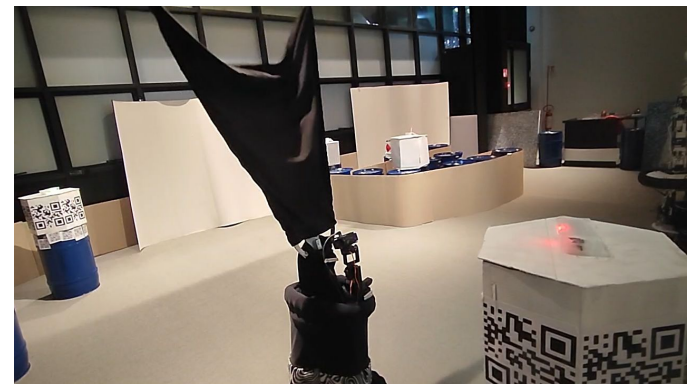


Objective: unlock the correct stations
avoiding the wrong ones

Controller's POV



Visitor's POV



The Stations - Virtual Maze

Goal oriented
activity

Visitor

Human
Translation

Goal oriented
activity

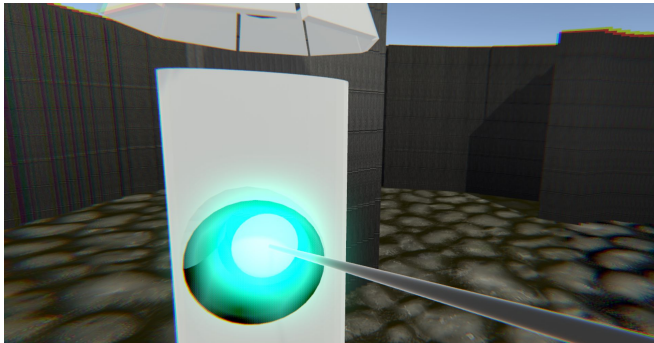
Physical
Avatar

Sensory
Translation
System
Controls

Controller

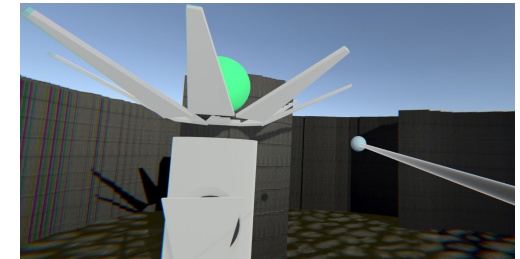
The **virtual** Stations

the virtual staff's **ball blinks**



the **joystick vibrates**

Correct Station



Wrong Station



The Stations - Physical Maze

Goal oriented
activity

Visitor

Human
Translation

Goal oriented
activity

Physical
Avatar

Sensory
Translation
System
Controls

Controller

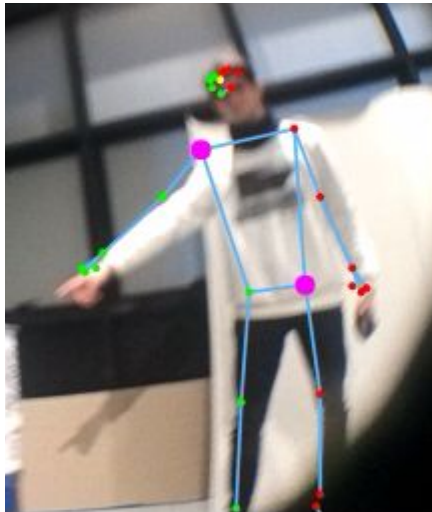
The **physical** Stations



Wrong and **correct**
Stations

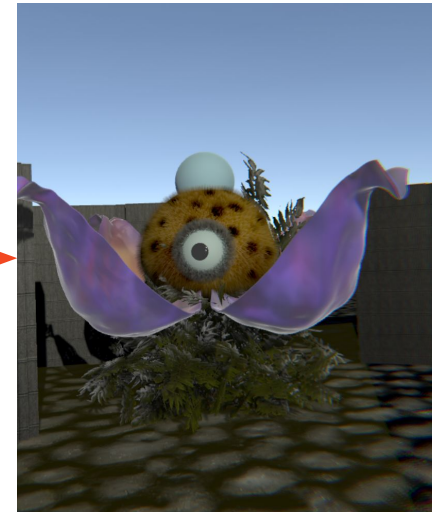
Physical Avatar **interaction** feedback





Human pose

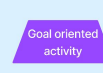
Translation of the
human pose



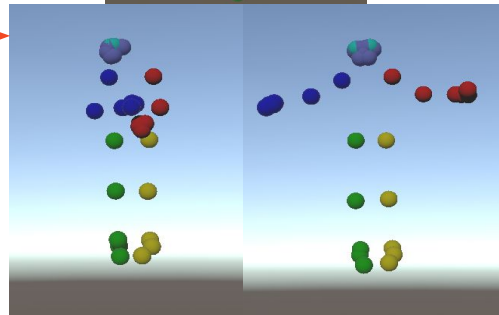
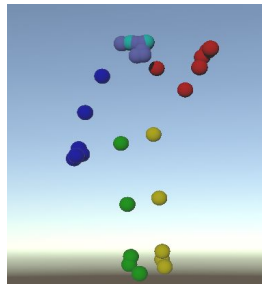
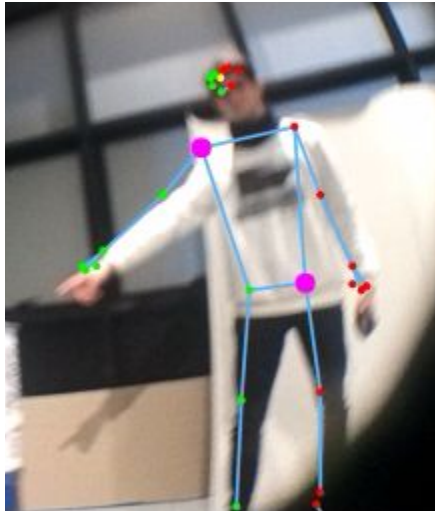
Non humanoid Entity

Human Translation - continued

Human Translation

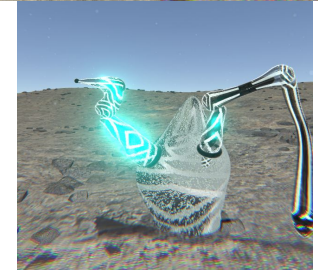
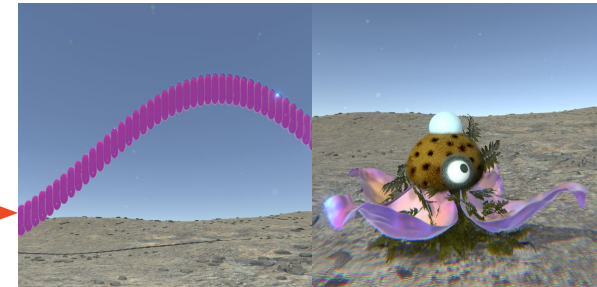


What the **camera** sees



3D pose information

Human Translations



Experimental setup

- The Mirrors

Human
Translation

Visitor

Human
Translation

Goal oriented
activity

Physical
Avatar

Sensory
Translation
System
Controls

Controller

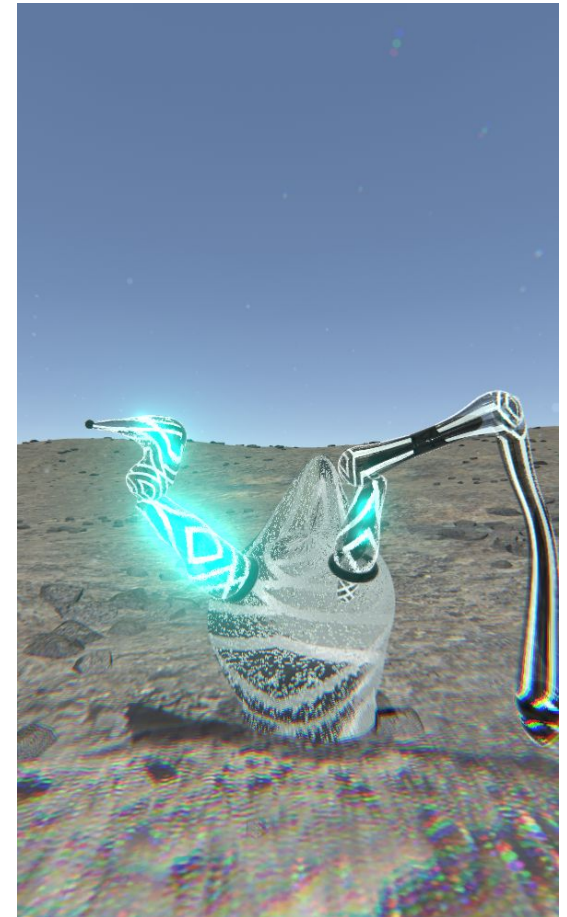
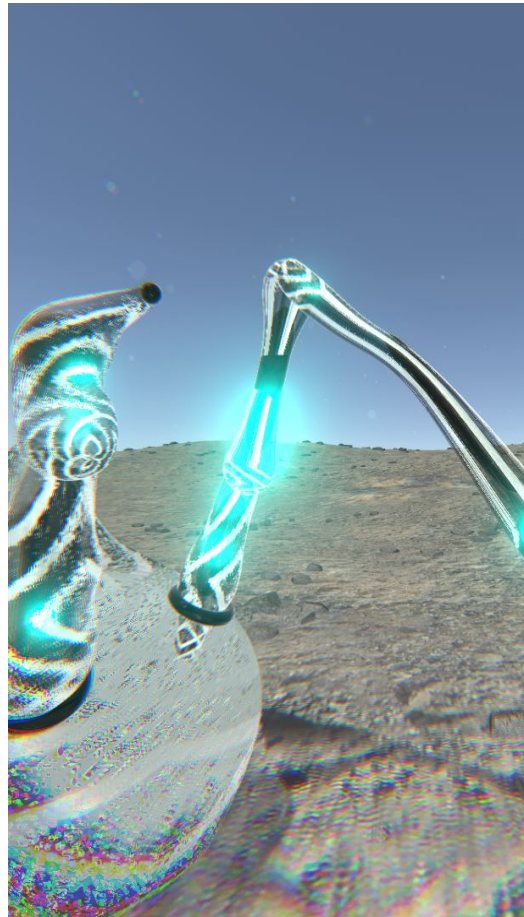
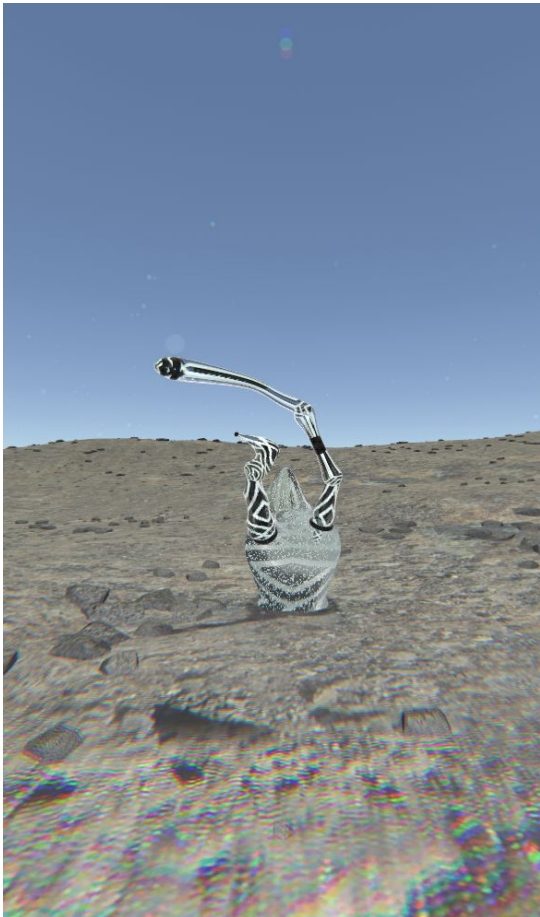
The Mirrors



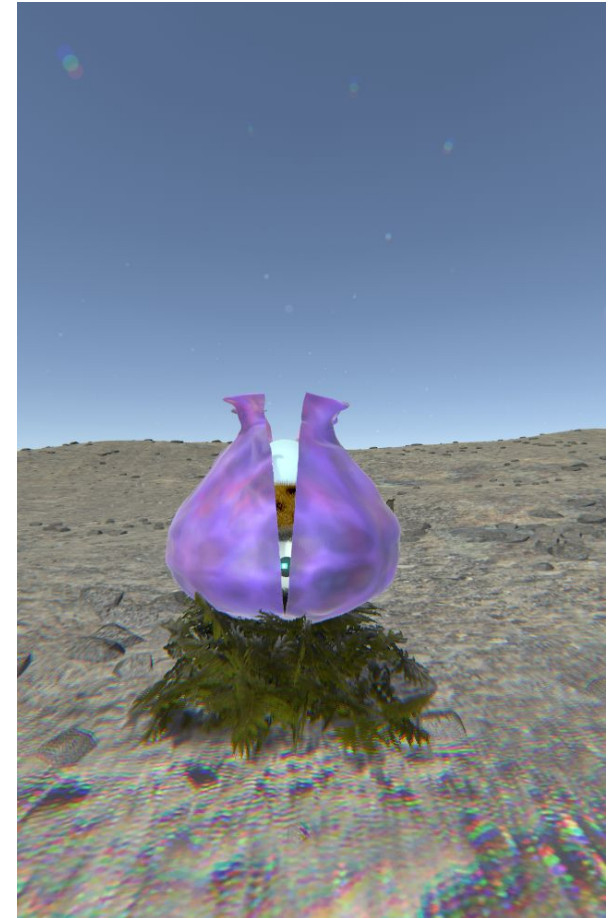
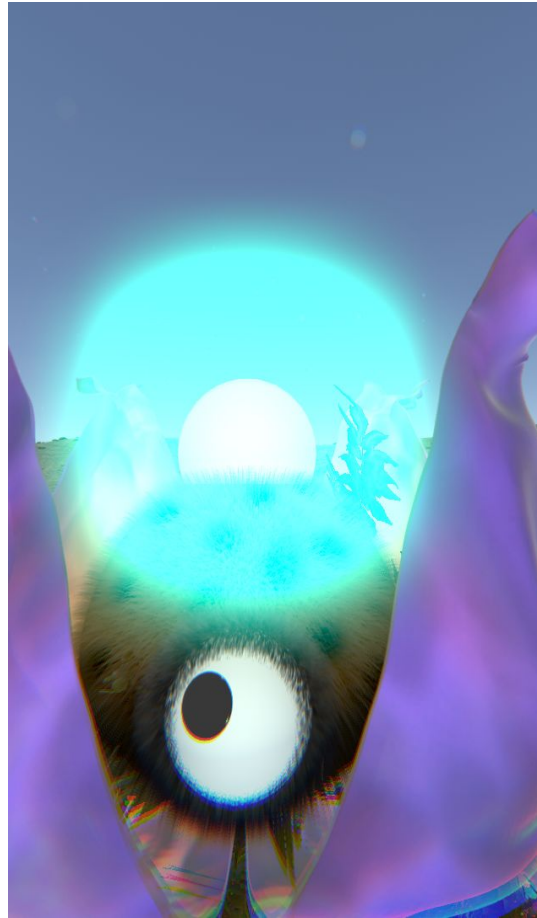
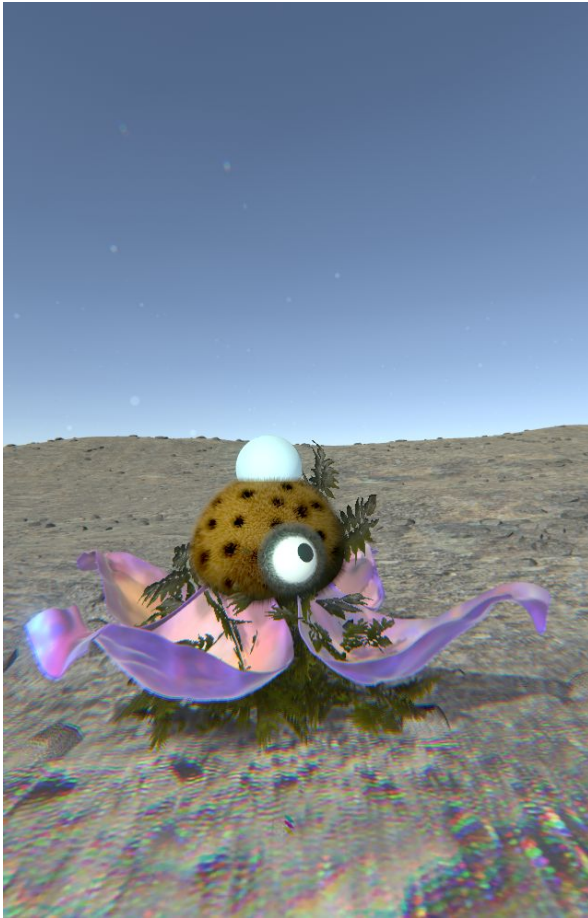
Ultrasonic sensor



Odile



Siid



Evangelion Human Translation

Human
Translation

Visitor

Human
Translation

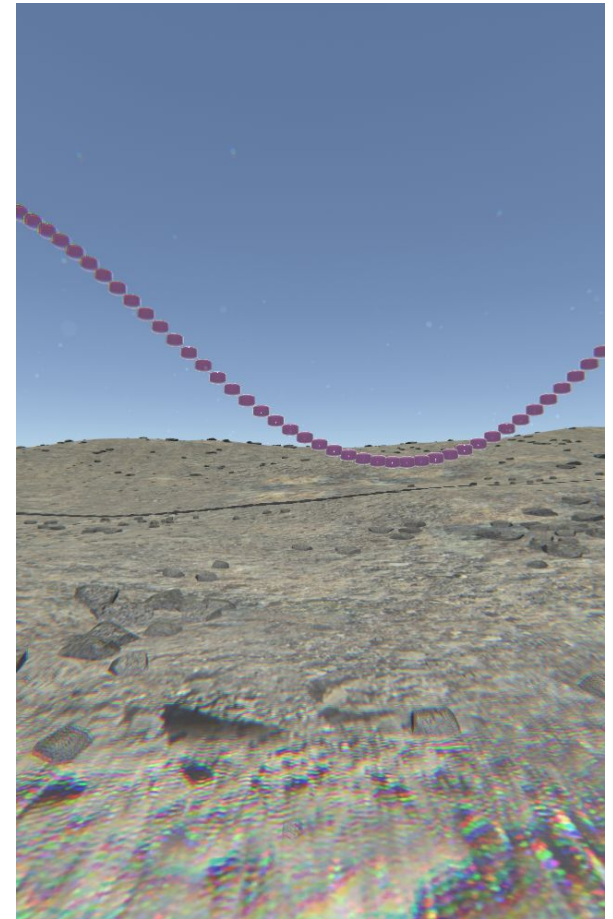
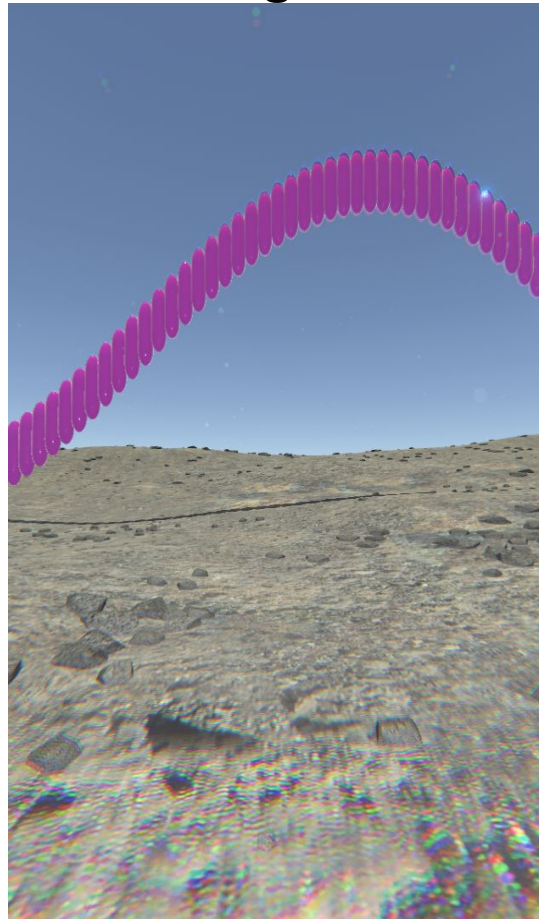
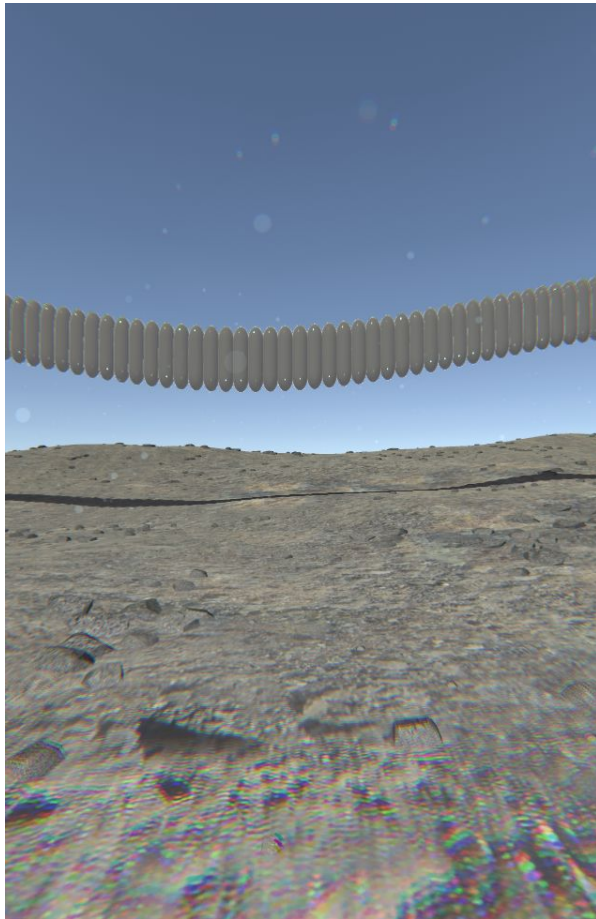
Goal oriented
activity

Physical
Avatar

Sensory
Translation
System
Controls

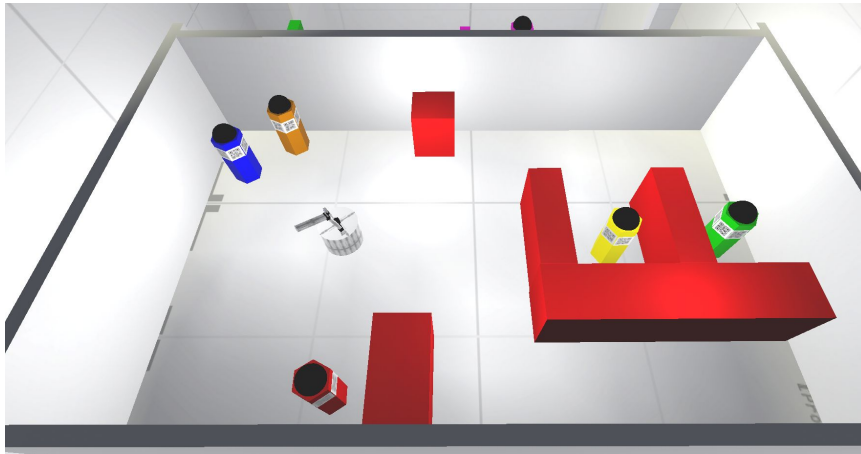
Controller

Evangelion

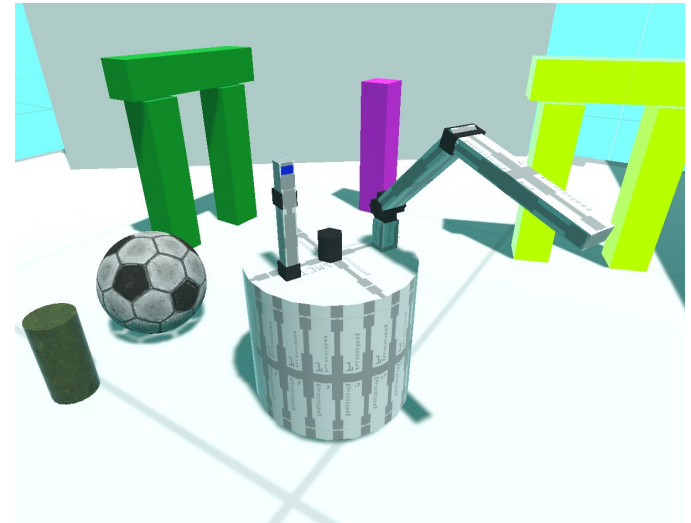




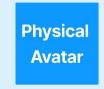
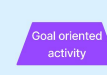
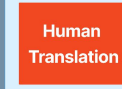
Maze simulation



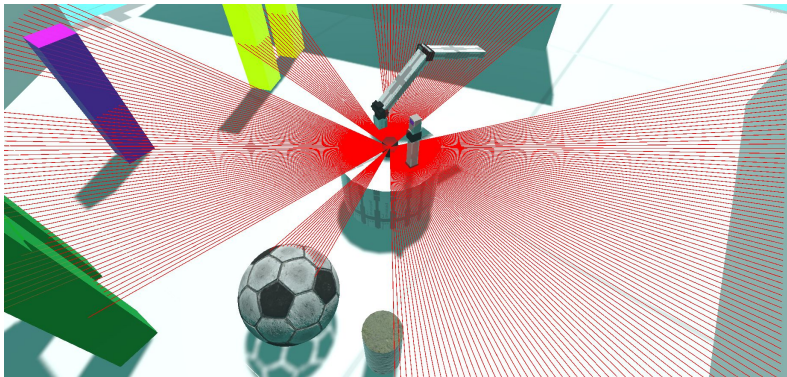
The **Virtual Avatar** - Virtual Odile



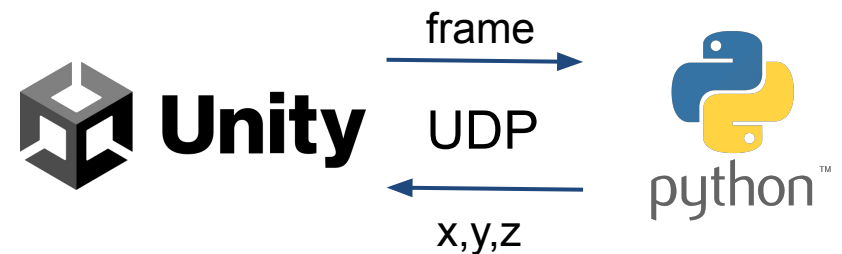
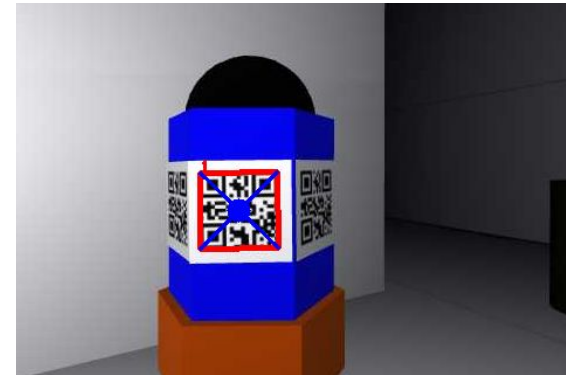
The Simulation - continued

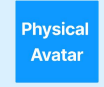
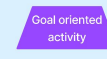
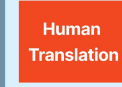


LIDAR simulation



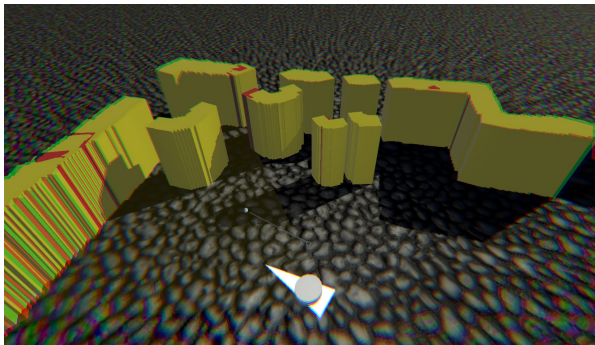
Camera simulation



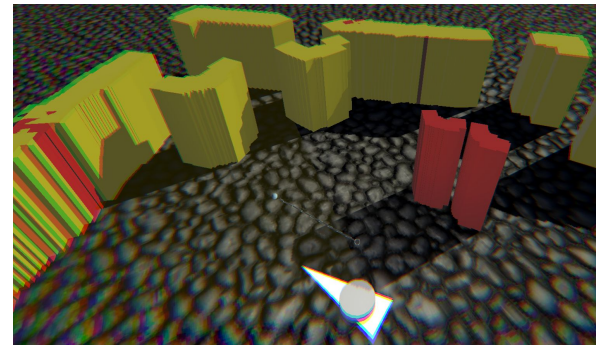


LIDAR information fusion

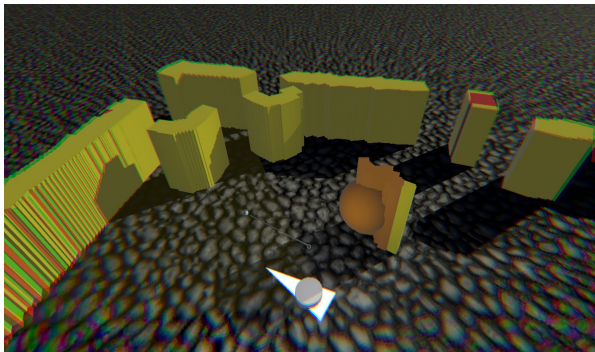
just noise



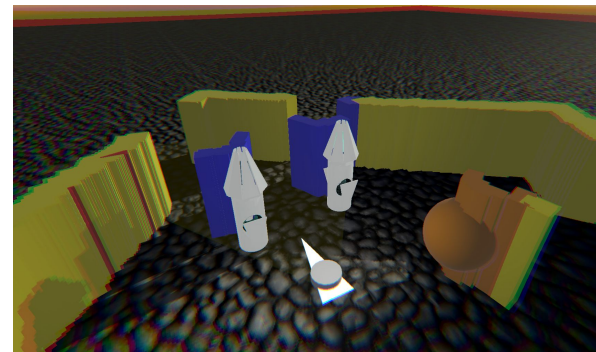
A person
moves



The **Person**
has been
tracked



The **Stations**
have been
detected



The experiments - the setup



Human
Translation

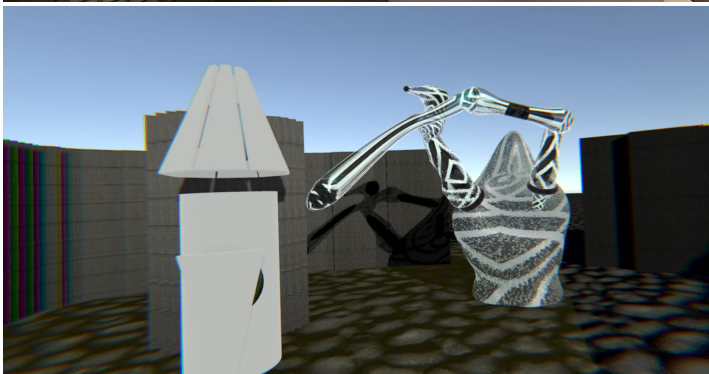
Goal oriented
activity

Physical
Avatar

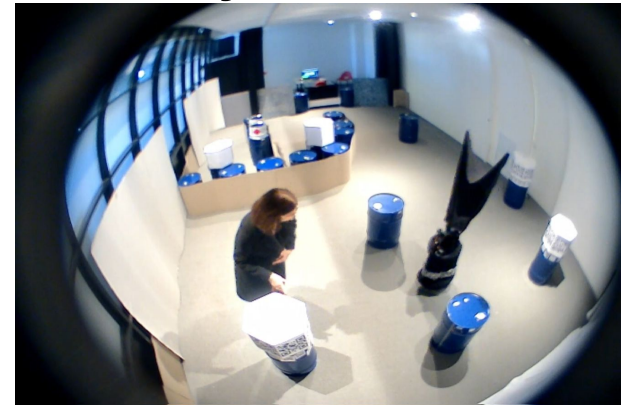
Sensory
Translation
System
Controls

Controller

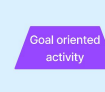
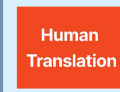
Virtual Maze



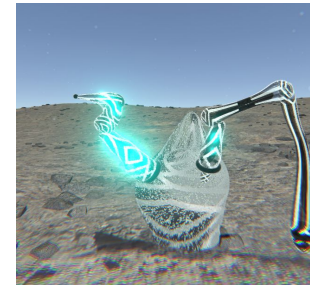
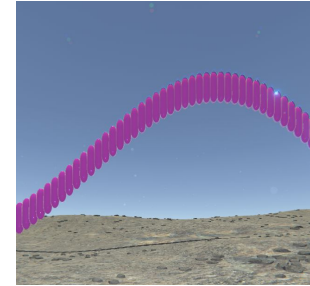
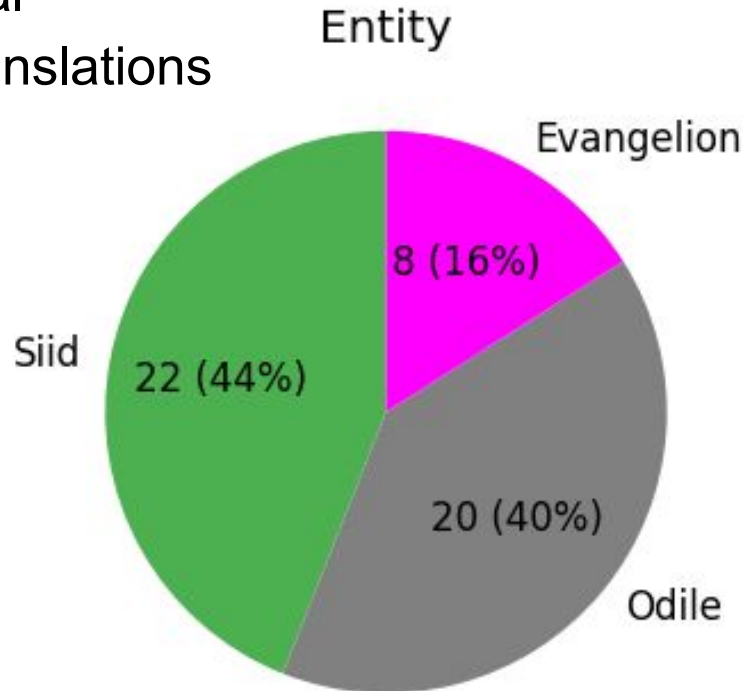
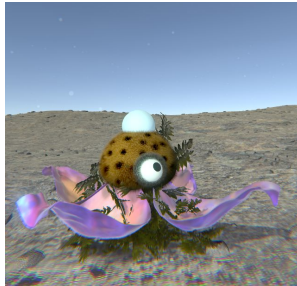
Physical Maze



The experiments - the quantities

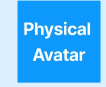
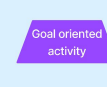
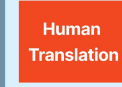


50 experiments in total
3 different Human Translations
(Entities)



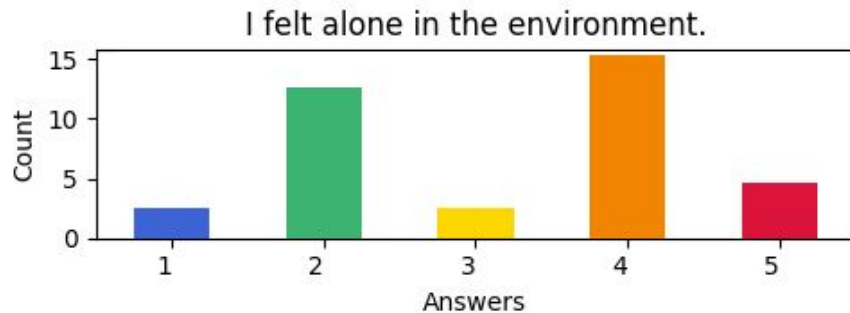
The experiments

- loneliness improvement

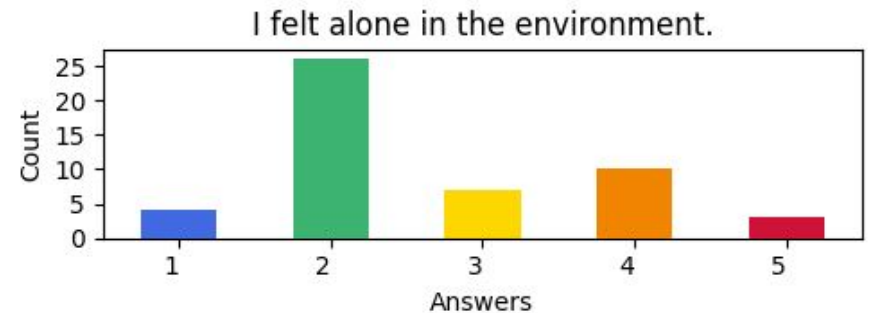


Feeling of **loneliness** in the virtual Maze

Previous system

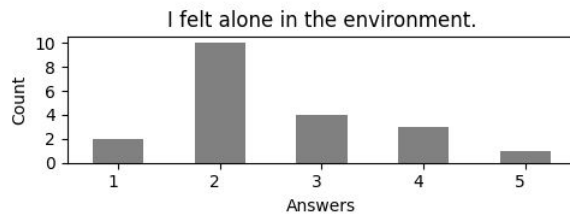


Current system

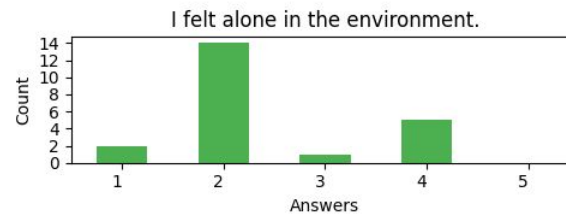


Current system's results by Human Translation

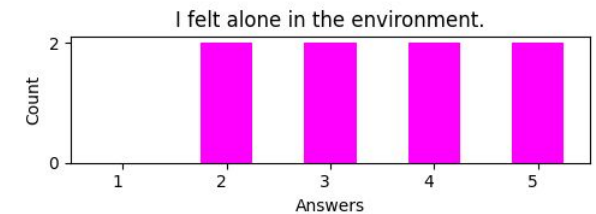
Odile



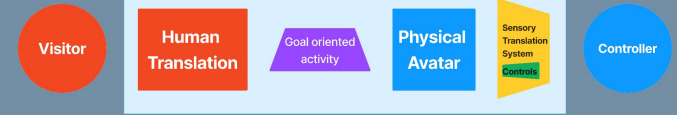
Siid



Evangelion

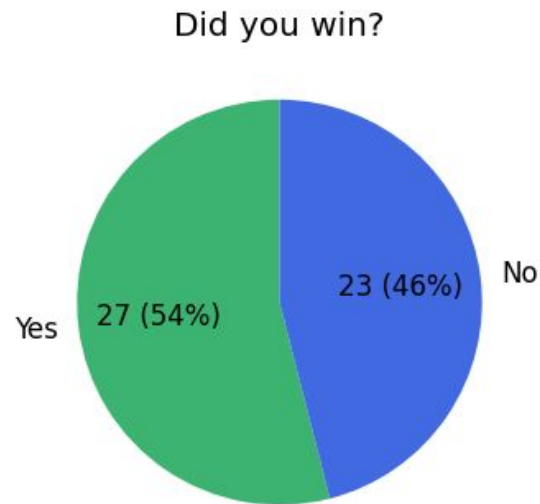
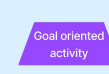
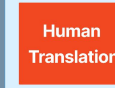


The experiments - our questions

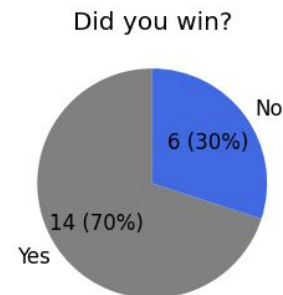
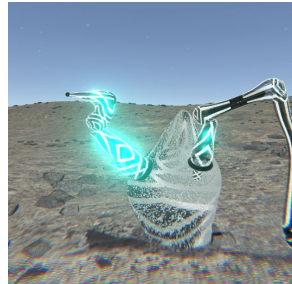


1. Were the two participants able to understand each other?
2. Were the two participants able to work together?

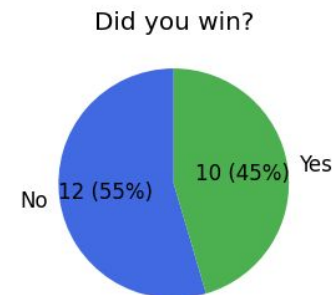
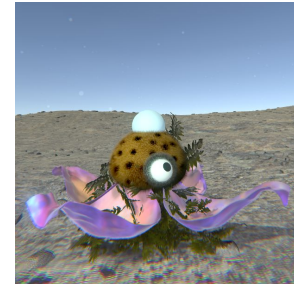
The experiments - win/loss



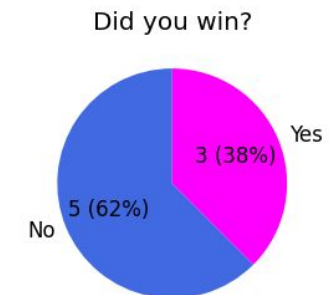
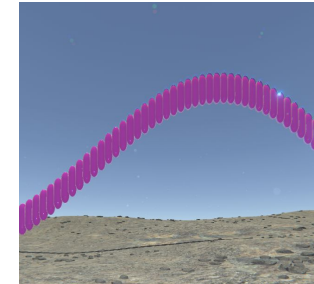
Odile



Siid

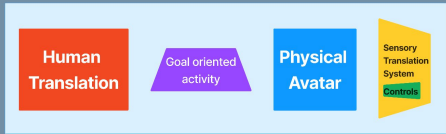


Evangelion

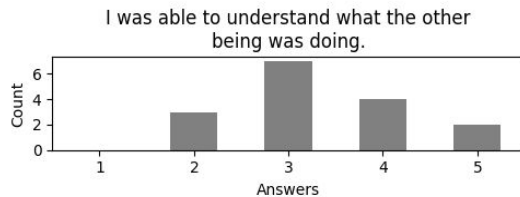
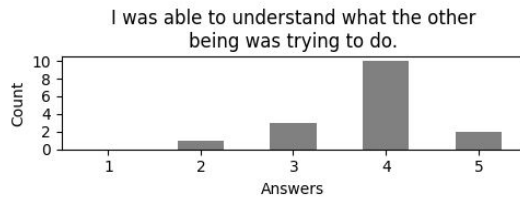
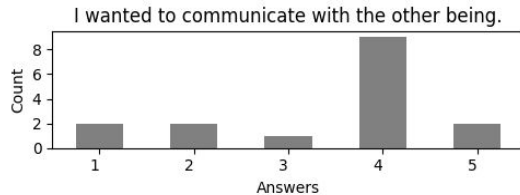
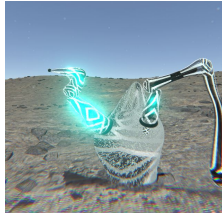


The experiments

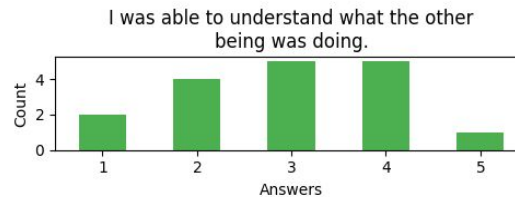
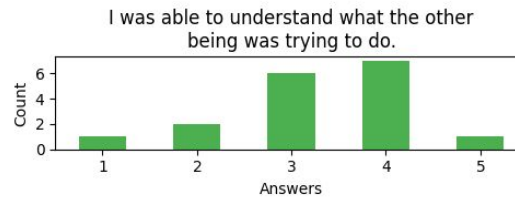
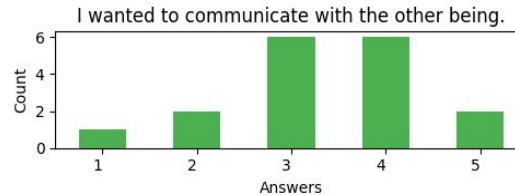
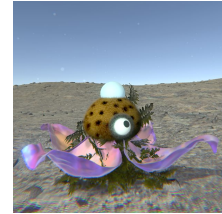
- communication with the Entity



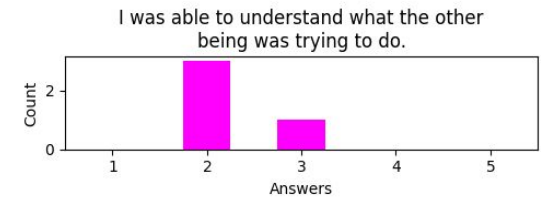
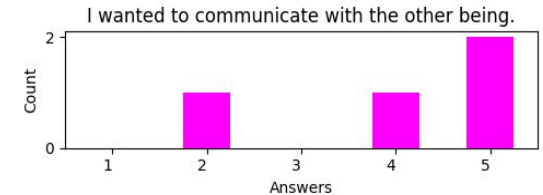
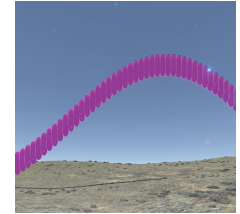
Odile



Siid

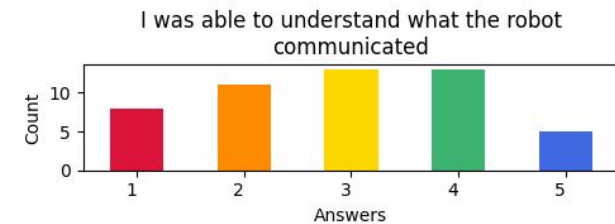
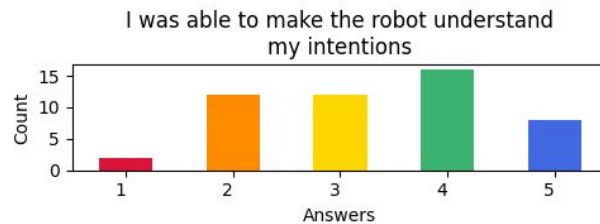
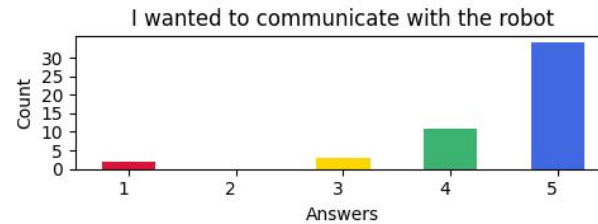
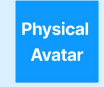
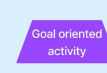
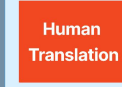


Evangelion

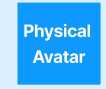
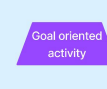
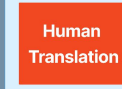


The experiments

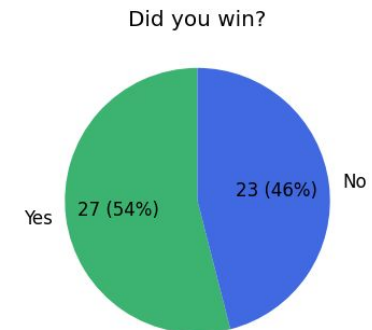
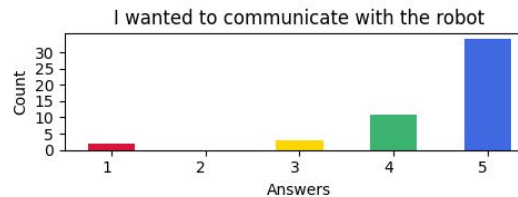
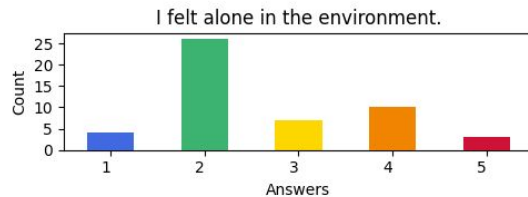
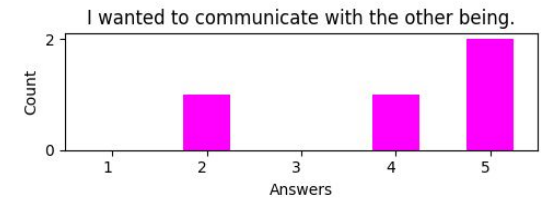
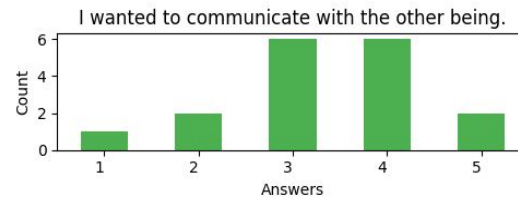
- communication with the avatar



The experiments - conclusions

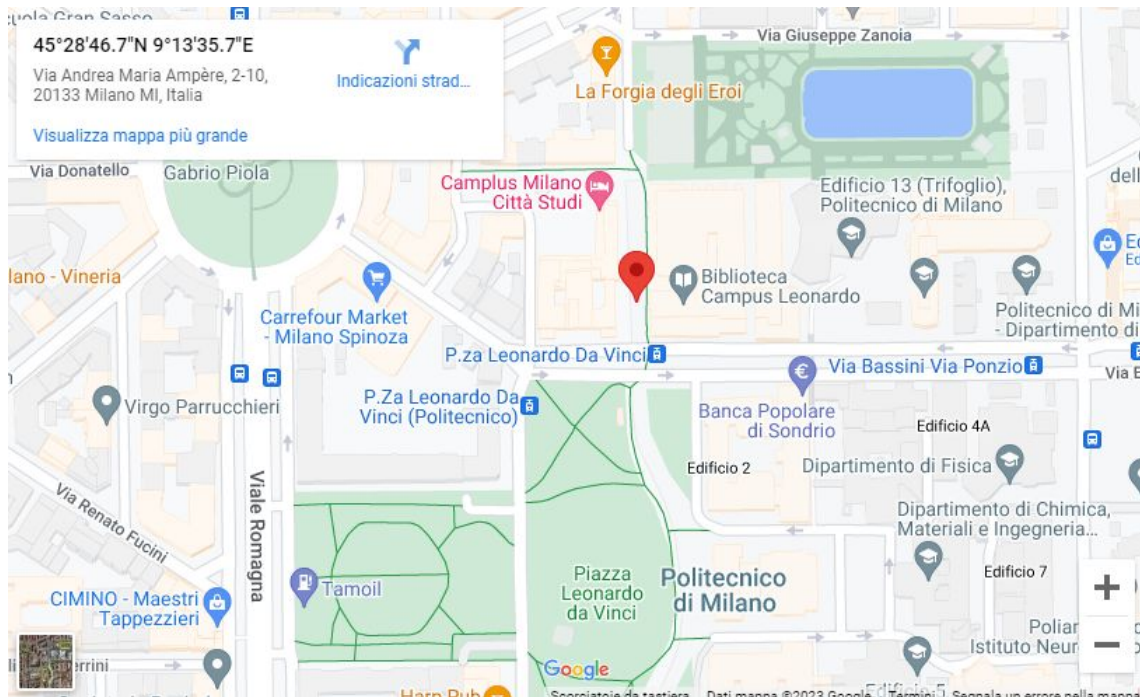


1. Were the two participants able to understand each other?
2. Were the two participants able to work together?



X-Cities exhibition

You can try all this and more at the X-Cities exhibition here at Politecnico di Milano in Spazio Mostre Guido Nardi, Via Ampère, 2, the show is open until 12th of January 2024 :-)



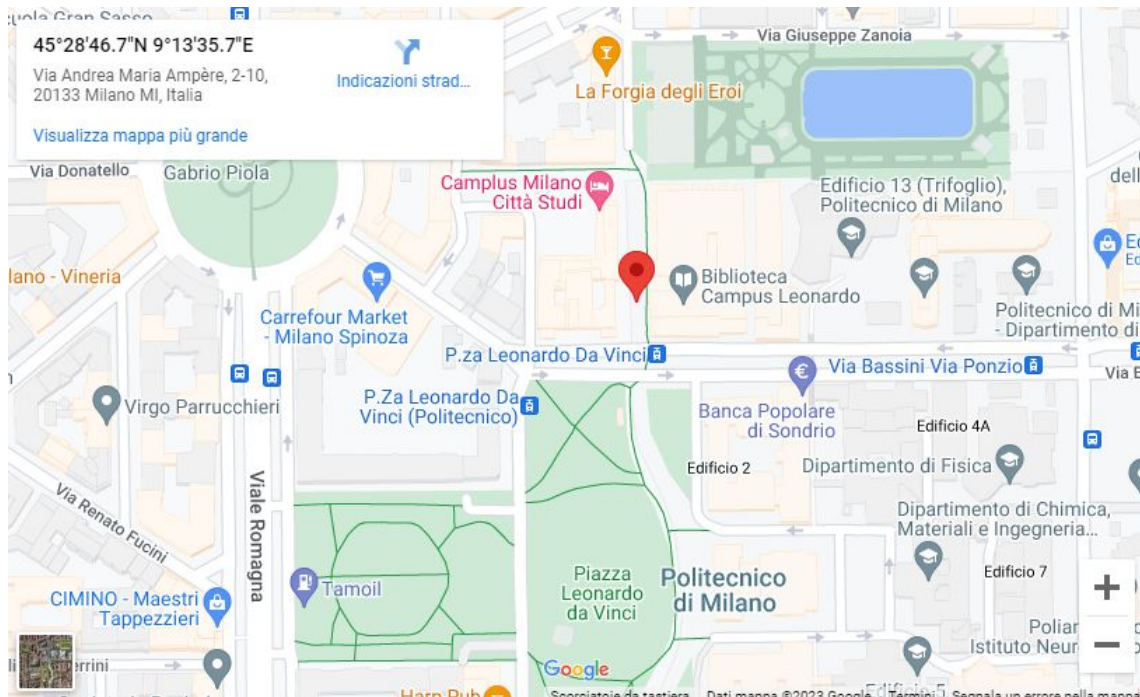
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That's it, thank you for your attention!



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POLITECNICO
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The diagram illustrates a human-machine interface system architecture. It features four main components: 'Human Translation' (orange rectangle), 'Physical Avatar' (blue rectangle), 'Sensory Translation System' (yellow trapezoid), and 'Goal oriented activity' (purple trapezoid). The 'Sensory Translation System' component contains a sub-component labeled 'Controls' (green rectangle). The components are arranged in a flow from left to right, with 'Human Translation' and 'Physical Avatar' at the top, and 'Goal oriented activity' and 'Sensory Translation System' at the bottom. A decorative horizontal line with vertical tick marks is located at the top of the slide.

**Human
Translation**

**Physical
Avatar**

**Sensory
Translation
System**

Controls

**Goal oriented
activity**